

Prob / Stats

MA2020S

Assigning probabilities

$R \leftarrow$ random expt.

$\Omega \leftarrow$ sample space, E - event of interest

Relative freq. $f_E = \frac{\# E \text{ occurs}}{n}$

where $n \leftarrow$ no. of replications of R .

$$p_E = \lim_{n \rightarrow \infty} f_E$$

Observation: Only for certain "key events" we need to assign probability. Then the probability of other events is automatically assigned.

R : random expt. , Ω : sample space

Expects: Collection of events should be "closed"
under complementation and countable union &
intersection of events.

Definition: (σ -Algebra)

A collection of sets $\mathcal{F}$ is called σ -algebra

if

- i) $A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}$
- ii) $A_1, A_2, \dots \in \mathcal{F}$, then
- $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$ and $\bigcap_{i=1}^{\infty} A_i \in \mathcal{F}$
- Imp*

Ex: R : Tossing a coin

$$\Omega = \{H, T\}$$

$$\mathcal{F} = 2^{\Omega} = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}$$

Ex: R : Random expt. with finite outcomes

$$\Omega = \{1, 2, \dots, n\}$$

$$\mathcal{F} = 2^{\Omega}$$

Ex: R : random expt. with $[0,1]$ as outcomes

$$\Omega = [0,1]$$

Note : All the intervals in Ω of the type

$$(a, b] \quad \text{for } a, b \in [0,1]$$

(Key events to which probability assignment can be done).

$$a_n = 1 - \frac{1}{n} \quad \text{for } n \in \mathbb{N}$$

$$A_n = (a_n, 1] \quad \bigcap_{n=1}^{\infty} A_n = \{1\}$$

Generate a σ -algebra $\mathcal{F} = \mathcal{B}[0,1]$ from these key intervals. (Borel - σ -algebra)

Definition :

Let Ω be the sample space for the random expt. R.

Let $\mathcal{F}$ be the σ -algebra associated with Ω .

Then the probability (measure) P is a function on $\mathcal{F}$ to $\mathbb{R}$ ($P: \mathcal{F} \rightarrow \mathbb{R}$) such that

i) $P(A) \geq 0 \quad \forall A \in \mathcal{F}$

ii) $P(\Omega) = 1$

iii) If A_i are mutually disjoint for $i=1, 2, \dots$
($A_i \cap A_j = \emptyset$ for $i \neq j$; $i, j=1, 2, \dots$), then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Properties:

$$i) P(A^c) = 1 - P(A)$$

observe: $A \cup A^c = \Omega$

$\uparrow$
disjoint

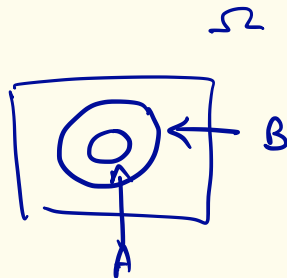
$$P(A \cup A^c) = P(A) + P(A^c) = P(\Omega) = 1$$

$$\Rightarrow P(A^c) = 1 - P(A)$$

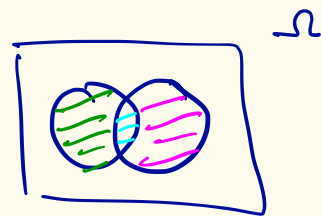
$$ii) P(\emptyset) = 0$$

$$iii) A \subseteq B \Rightarrow P(A) \leq P(B)$$

$$B = A \cup (B \cap A^c)$$



$$4) P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



$$A \cup B = (A \cap B^c) \cup (A^c \cap B) \cup (A \cap B)$$

$$P(A \cup B) = P(A \cap B^c) + P(A^c \cap B) + P(A \cap B) \quad \text{--- (1)}$$

$$A = (A \cap B^c) \cup (A \cap B)$$

$$P(A) = P(A \cap B^c) + P(A \cap B) \quad \text{--- (2)}$$

$$P(B) = P(A^c \cap B) + P(A \cap B) \quad \text{--- (3)}$$

$$5) P(A_1 \cup A_2 \cup A_3) = P(A_1) + P(A_2) + P(A_3) - P(A_1 \cap A_2) - P(A_2 \cap A_3) - P(A_1 \cap A_3) + P(A_1 \cap A_2 \cap A_3)$$

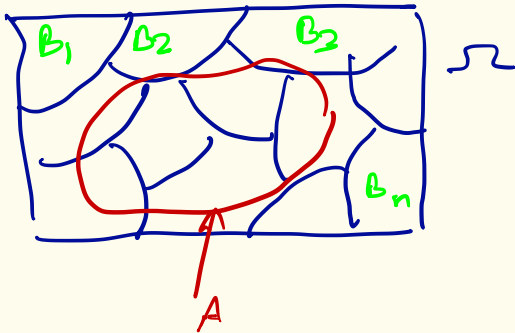
Disjoint cover of Ω

Let $\{B_1, B_2, \dots, B_n\}$ be a collection (subset of $\mathcal{F}$)
be such that

i) $B_i \cap B_j = \emptyset$ for $i \neq j$, $i, j = 1, 2, \dots, n$

ii) $\bigcup_{i=1}^n B_i = \Omega$

Then $\{B_1, \dots, B_n\}$ is called as disjoint cover of Ω .



For any $A \in \mathcal{F}$

$$P(A) = P(A \cap B_1) + P(A \cap B_2) + \dots + P(A \cap B_n)$$